

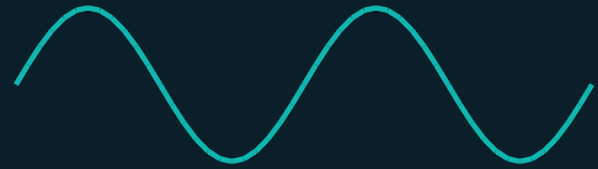
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SIGNAL INTELLIGENCE WORKSTATION

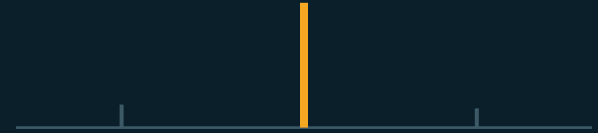
A hybrid MATLAB DSP + Python machine-learning signal analyzer for audio and image inputs

PROJECT TEAM

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sinusoidal



impulse



step

01

PROBLEM & SCOPE

Why classify a signal before analysing it?

A one-size-fits-all DSP recipe cannot suit every kind of signal.

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THE CHALLENGE

Real audio and image inputs arrive noisy, in different formats, sampling rates and sizes, and their signal type is not known in advance.

OUR APPROACH

Standardise the input in MATLAB, identify the signal class with a Python ML model, then run only the DSP analyses that suit that class.

PROJECT OBJECTIVES

1

Standardise inputs

16 kHz / 32,000-sample audio;
128×128 image

2

Classify the signal

Impulse, sinusoidal, white noise, step, chirp

3

Route to class-aware DSP

FFT, STFT, wavelets, FIR, IIR, conv, deconv

4

Present unified results

One result structure plus visualisations

02

METHODOLOGY

End-to-end pipeline architecture

Data moves from MATLAB to Python in memory as a numpy array.

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Import signal

Raw audio file or raw image is loaded by the pipeline

2

MATLAB preprocessing

Audio: mono, DC removal, 16 kHz, normalise, 32,000 samples. Image: grayscale, 128×128, [0, 1]

3

Python ML model

Audio: 35 features → scaler → .pkl model. Image: 128×128 array → CNN .pkl model

4

Prediction

Returns class, class_id, confidence and 5-class probabilities

5

MATLAB DSP analysis

Class-aware FFT, STFT, wavelets, FIR, IIR, convolution, deconvolution → final results

03

MATLAB STAGE

Preprocessing: one fixed format per modality

Every input is conditioned before it reaches the ML model.

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16 kHz

Unified sampling rate

32,000

Samples per clip (2.0 s)

128×128

Grayscale image size

AUDIO

- Stereo to mono
- DC offset removal
- Resample to 16,000 Hz
- Normalise amplitude to $[-1, 1]$
- Fix length to 32,000 samples (class-aware crop / pad)

Output: $32,000 \times 1$ vector

IMAGE

- RGB / RGBA to grayscale (1 channel)
- Resize to exactly 128×128
- Min-max normalise pixels to $[0.0, 1.0]$
- Original size and type kept as metadata

Output: $128 \times 128 \times 1$ array

04

MACHINE LEARNING

Python classifier: five signal classes

The prediction decides which DSP
analyses run next.

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0

impulse

1

sinusoidal

2

white_noise

3

step

4

chirp

AUDIO MODEL

- 32,000-sample vector
- 35 extracted features
- Feature scaler
- .pkl classifier

IMAGE MODEL

- 128 × 128 array
- Passed directly to CNN
- .pkl model
- 5-class probabilities

PREDICTION RETURNED TO MATLAB

<code>class</code>	predicted signal label
<code>class_id</code>	integer 0 to 4
<code>confidence</code>	probability of the top class
<code>probabilities</code>	score for each of the 5 classes

05

DSP ROUTER

Class-aware DSP analysis

Preprocessed signal + predicted
class decide the analyses.

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Class	FFT	STFT	Wavelet	FIR	IIR	Conv	Deconv
Impulse	●	•	●	○	•	○	○
Sinusoidal	●	•	•	○	○	•	•
White noise	●	•	•	○	○	•	•
Step	●	•	●	○	○	○	•
Chirp	●	●	●	•	•	•	○

● primary ○ secondary

IMAGE PATH: 2-D COUNTERPARTS

- 2-D FFT and image filtering run for every class
- 2-D wavelet is added for impulse, step and chirp
- Secondary: 2-D convolution (impulse, sinusoidal); 2-D deconvolution (impulse, chirp)

06

AUDIO OUTPUT

Audio pipeline result

Sample audio through all four
pipeline stages.

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Field	Result
Input file	<i>(from report)</i>
Preprocessed	32,000 samples @ 16 kHz
Predicted class	<i>(from report)</i>
ML confidence	<i>(from report)</i>
FFT peak	<i>(from report)</i>
FIR / IIR filter	<i>(from report)</i>
DSP status	<i>(from report)</i>

07

IMAGE OUTPUT

Image pipeline result

Sample image through all four pipeline stages.

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Field	Result
Input file	(from report)
Preprocessed	128 × 128, double, [0, 1]
Predicted class	(from report)
ML confidence	(from report)
2-D FFT	(from report)
Image filter	(from report)
DSP status	(from report)

08

VERIFICATION

10-case end-to-end test

One audio and one image sample per class.

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Type	Input class	Primary DSP	Predicted	Conf.
Audio	impulse	FFT, Wavelet	<i>(report)</i>	<i>(report)</i>
Audio	sinusoidal	FFT	<i>(report)</i>	<i>(report)</i>
Audio	white_noise	FFT	<i>(report)</i>	<i>(report)</i>
Audio	step	FFT, Wavelet	<i>(report)</i>	<i>(report)</i>
Audio	chirp	FFT, STFT, Wavelet	<i>(report)</i>	<i>(report)</i>
Image	impulse	2-D FFT, 2-D Wavelet, Filter	<i>(report)</i>	<i>(report)</i>
Image	sinusoidal	2-D FFT, Filter	<i>(report)</i>	<i>(report)</i>
Image	white_noise	2-D FFT, Filter	<i>(report)</i>	<i>(report)</i>
Image	step	2-D FFT, 2-D Wavelet, Filter	<i>(report)</i>	<i>(report)</i>
Image	chirp	2-D FFT, 2-D Wavelet, Filter	<i>(report)</i>	<i>(report)</i>

09

CONCLUSION

What we achieved

Integrated MATLAB and Python, end to end.

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01

End-to-end pipeline

Audio and image inputs flow from raw file to DSP results across MATLAB and Python.

02

Standardised preprocessing

Every input becomes 16 kHz / 32,000 samples or a 128×128 image.

03

Class-aware DSP

Analyses are chosen from the predicted class, not one fixed recipe.

04

Verified on 10 cases

Five audio and five image samples, one per class, tested end to end.

NEXT STEPS

- Test on larger, noisier datasets
- Report noise-extraction quality, e.g. SNR improvement
- Support real-time / streaming input
- Add more signal classes

Thank You

SPANDHAN: Noise Extraction from a Noisy Signal

Dorthi Raj • Divyansh Awasthi • Gaurab Gupta • Gaurav Dubey

Questions are welcome